

Zeren Shen

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TECHNICAL SUMMARY

LLM & AI Engineer with 3+ years of experience building production-grade Generative AI systems. Currently at Unilever, developing enterprise agentic AI applications that combine LLMs, RAG, tool calling, data analytics, and evaluation. Experienced across the AI stack from model fine-tuning and inference optimization to backend services and end-to-end AI applications.

PROFESSIONAL EXPERIENCE

LLM Engineer

May 2025 - Present

Unilever — Horizon3 AI Labs

Toronto, Canada

- Build production-grade agentic AI systems across procurement, sustainability, tariff intelligence, and negotiation, leveraging Google Gemini, Google ADK, and LangGraph.
- Design LLM workflows integrating RAG, tool calling, web research, BigQuery/SQL analytics, and structured outputs, enabling natural-language access to enterprise data and automated business analysis.
- Develop end-to-end AI applications with Python, FastAPI, React/Streamlit, GCP, BigQuery, and MongoDB/Cosmos DB, integrating agent orchestration, APIs, enterprise data, and production services.
- Develop LLM evaluation and reliability workflows using DeepEval and LLM-as-a-judge methods to assess correctness, completeness, relevance, and agent performance.

Machine Learning Engineer

May 2023 - Nov 2024

ThinkGenAI Lab Inc.

Toronto, Canada

- Built an end-to-end RAG system for research teams, automating PDF ingestion, semantic retrieval, and LLM-based summarization using LangChain, Pinecone, FastAPI, and Docker; implemented ReAct agents for real-time research and question answering.
- Developed a real-time voice AI pipeline integrating Whisper STT, GPT-3.5, and Google TTS, achieving 1.2s end-to-end latency; optimized asynchronous processing, audio-chunk pipelining, request batching, and token-level streaming to support 20+ requests/min and reduce response latency by 50%.
- Developed deep learning systems for 3D human motion understanding and generation, including a conditional diffusion model with MANO achieving 2.8mm MPJPE and a Transformer-based golf posture classifier achieving 92% F1.

SELECTED LLM PROJECT

Medical Domain LLM Fine-tuning

July 2024 - Jan 2025

- Fine-tuned LLaMA-3-8B on PubMedQA using QLoRA, 4-bit quantization, and DeepSpeed ZeRO-3 for scalable multi-GPU training on Lambda Cloud.
- Improved clinical QA accuracy by 18.7% and deployed vLLM-based inference with PagedAttention for low-latency generation.

EDUCATION

University of Toronto

Sep 2021 - Nov 2022

Master of Engineering, Mechanical and Industrial Engineering - AI/ML Focus

University of Waterloo

Sep 2016 - Apr 2021

Bachelor of Mathematics, Statistics & Machine Learning, Minor in Computer Science

KEY SKILLS

- **GenAI:** Agentic AI, Tool Calling, LLM Evaluation, Prompt Engineering, LangChain, LangGraph, GoogleADK
- **AI Engineering:** PyTorch, Python, FastAPI, React, Streamlit, REST APIs, WebSockets, AsyncIO
- **Cloud / Infrastructure:** GCP, BigQuery, AWS, Azure, Docker, MongoDB, Cosmos DB